HW8

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Q1: 4.3

Let ν be the counting measure on $[0,1]$ and $\int_{[0,1]} h\,d\nu < \infty$ for some nonnegative function h. Show that $B = \{x : h(x) > 0\}$ is countable.

(**Hint:** Let $B_n = \{x : h(x) > \frac{1}{n}\}$. Show that B_n is a finite set for each $n \in \mathbb{N}$.)

Answer

We utilize the following results:

- Counting measure: $\nu(A)$ is the number of elements of A; so $\nu(A) < \infty$ iff A is finite.
- Corollary 2.3.3: Proposition 2.3.1 holds for nonnegative measurable functions; in particular, $\int \alpha f\,d\mu = \alpha \int f\,d\mu$ for $\alpha \geq 0$, and $f \geq g$ a.e. (μ) implies $\int f\,d\mu \geq \int g\,d\mu$.
- Proposition A.2.1 (Axiom of Archimedes): for every $y \in \mathbb{R}$, there exists $n \in \mathbb{N}$ with $n > y$.

Proof. (h is taken to be measurable, as the integral presumes.) For $n \in \mathbb{N}$, define $B_n \equiv \{x \in [0,1] : h(x) > \frac{1}{n}\}$, a measurable set since h is measurable.

B_n is finite: Let $n \in \mathbb{N}$ be given. On B_n, $h > \frac{1}{n}$, and off B_n, $h \geq 0$, so $h \geq \frac{1}{n} I_{B_n}$ on $[0,1]$. By Corollary 2.3.3,

$$\frac{1}{n}\,\nu(B_n) = \int \frac{1}{n} I_{B_n}\,d\nu \leq \int_{[0,1]} h\,d\nu < \infty.$$

Thus $\nu(B_n) < \infty$, i.e., B_n is finite.

B is countable: If $x \in B$, then $h(x) > 0$, and by Proposition A.2.1 there is an $n \in \mathbb{N}$ with $n > 1/h(x)$, i.e., $h(x) > \frac{1}{n}$, so $x \in B_n$. Conversely, each $B_n \subset B$. So $B = \bigcup_{n \geq 1} B_n$, a countable union of finite sets, and thus countable. //

Said differently, this is the fact that makes Remark 4.1.1 work, where it shows the Radon-Nikodym theorem can fail when ν is not σ-finite.

Q2: 4.4

Prove Proposition 4.1.2.

Proposition 4.1.2: Let $\nu, \mu, \mu_1, \mu_2, \ldots$ be σ-finite measures on a measurable space $(\Omega, \mathcal{F})$.

Answer

We utilize the following results:

- Definition 4.1.1 (absolute continuity): $\mu \ll \nu$ if $\nu(A) = 0 \Rightarrow \mu(A) = 0$ for all $A \in \mathcal{F}$.
- Definition 4.1.2 (singularity): $\mu \perp \nu$ if there is a $B \in \mathcal{F}$ with $\mu(B) = 0$ and $\nu(B^c) = 0$.
- Definition 4.1.3 (Radon-Nikodym derivative): a nonnegative measurable h with $\mu(A) = \int_A h\,d\nu$ for all $A \in \mathcal{F}$ is written $\frac{d\mu}{d\nu} = h$; when μ is σ-finite, it is unique a.e. (ν).
- Example 4.1.3: if $h \geq 0$ is measurable and $\mu(A) \equiv \int_A h\,d\nu$, then μ is a measure and $\mu \ll \nu$.

- Theorem 4.1.1 (Lebesgue-Radon-Nikodym): for σ-finite μ, ν, (i) $\mu = \mu_a + \mu_s$ uniquely, with $\mu_a \ll \nu$ and $\mu_s \perp \nu$; (ii) there is a nonnegative measurable h with $\mu_a(A) = \int_A h \, d\nu$ for all $A \in \mathcal{F}$. In particular, if $\mu \ll \nu$, then $\mu_a = \mu$, and $\frac{d\mu}{d\nu}$ exists.
- Definitions 2.3.2 and 2.3.3 (integrals of simple and of nonnegative measurable functions), with nonnegative simple $h_n \uparrow h$ as in (3.7).
- Corollary 2.3.3 (see Q1); in particular, $\int (\alpha f + \beta g) \, d\mu = \alpha \int f \, d\mu + \beta \int g \, d\mu$ for $\alpha, \beta \geq 0$.
- Theorem 2.3.4 (monotone convergence theorem, MCT): for nonnegative measurable $f_n \uparrow f$, $\int f_n \, d\mu \uparrow \int f \, d\mu$.
- Corollary 2.3.5: for nonnegative measurable $\{h_n\}_{n \geq 1}$, $\int \left(\sum_{n=1}^{\infty} h_n \right) d\mu = \sum_{n=1}^{\infty} \int h_n \, d\mu$.
- Problem 2.20: for nonnegative $\{a_{ij}\}$, $\sum_i \sum_j a_{ij} = \sum_j \sum_i a_{ij}$.

Lemma: Let μ, ν be measures on $(\Omega, \mathcal{F})$, and let $f \geq 0$ be measurable with $\mu(A) = \int_A f \, d\nu$ for all $A \in \mathcal{F}$. Then for every nonnegative measurable h,

$$\int h \, d\mu = \int hf \, d\nu. \tag{1}$$

Proof. We build up from indicators.

Indicator h: If $h = I_A$, $A \in \mathcal{F}$, then $\int I_A \, d\mu = \mu(A) = \int_A f \, d\nu = \int I_A f \, d\nu$.

Nonnegative simple h: If $h = \sum_{i=1}^{k} c_i I_{A_i}$ with $c_i \geq 0$, then by Corollary 2.3.3 and the indicator case,

$$\int h \, d\mu = \sum_{i=1}^{k} c_i \int I_{A_i} \, d\mu = \sum_{i=1}^{k} c_i \int I_{A_i} f \, d\nu = \int hf \, d\nu.$$

Nonnegative measurable h: Let $\{h_n\}_{n \geq 1}$ be nonnegative simple functions with $h_n \uparrow h$, as in (3.7). Then $h_n f \uparrow hf$, and by Definition 2.3.3, the simple case, and the MCT,

$$\int h \, d\mu = \lim_{n \to \infty} \int h_n \, d\mu = \lim_{n \to \infty} \int h_n f \, d\nu = \int hf \, d\nu. \quad //$$

Throughout, each Radon-Nikodym derivative used exists by Theorem 4.1.1, since all measures involved are σ-finite.

(i)

If $\mu_1 \ll \mu_2$ and $\mu_2 \ll \mu_3$, then $\mu_1 \ll \mu_3$ and

$$\frac{d\mu_1}{d\mu_3} = \frac{d\mu_1}{d\mu_2} \frac{d\mu_2}{d\mu_3} \quad \text{a.e. } (\mu_3).$$

Answer

Proof. $\mu_1 \ll \mu_3$**:** Suppose $\mu_3(A) = 0$ for some $A \in \mathcal{F}$. Then $\mu_2(A) = 0$, since $\mu_2 \ll \mu_3$, and so $\mu_1(A) = 0$, since $\mu_1 \ll \mu_2$. Thus $\mu_1 \ll \mu_3$.

The chain rule: Let $A \in \mathcal{F}$ be given. By (1), with μ_2, μ_3 and $\frac{d\mu_2}{d\mu_3}$ in the roles of μ, ν and f, and with $h = I_A \frac{d\mu_1}{d\mu_2}$,

$$\int_A \frac{d\mu_1}{d\mu_2} \frac{d\mu_2}{d\mu_3} \, d\mu_3 = \int_A \frac{d\mu_1}{d\mu_2} \, d\mu_2 = \mu_1(A).$$

So $\frac{d\mu_1}{d\mu_2} \frac{d\mu_2}{d\mu_3}$, being nonnegative and measurable, is a Radon-Nikodym derivative of μ_1 w.r.t. μ_3, and by uniqueness (Definition 4.1.3), $\frac{d\mu_1}{d\mu_3} = \frac{d\mu_1}{d\mu_2} \frac{d\mu_2}{d\mu_3}$ a.e. (μ_3). //

(ii)

Suppose that μ_1 and μ_2 are dominated by μ_3. Then for any $\alpha, \beta \geq 0$, $\alpha\mu_1 + \beta\mu_2$ is dominated by μ_3 and

$$\frac{d(\alpha\mu_1 + \beta\mu_2)}{d\mu_3} = \alpha\frac{d\mu_1}{d\mu_3} + \beta\frac{d\mu_2}{d\mu_3} \quad \text{a.e. } (\mu_3).$$

Answer

Proof. $\alpha\mu_1 + \beta\mu_2$ **is a σ-finite measure:** It is nonnegative and vanishes at $\emptyset$. For disjoint $A_1, A_2, \ldots \in \mathcal{F}$, adding two series of nonnegative terms termwise,

$$(\alpha\mu_1 + \beta\mu_2)\Big(\bigcup_{k\geq 1} A_k\Big) = \alpha\sum_{k=1}^{\infty}\mu_1(A_k) + \beta\sum_{k=1}^{\infty}\mu_2(A_k) = \sum_{k=1}^{\infty}(\alpha\mu_1 + \beta\mu_2)(A_k).$$

For σ-finiteness, take partitions $\{D_n\}$ and $\{E_m\}$ of Ω with $\mu_1(D_n) < \infty$ and $\mu_2(E_m) < \infty$. Then $\{D_n \cap E_m\}_{n,m\geq 1}$ is a countable partition of Ω on which $\alpha\mu_1 + \beta\mu_2$ is finite.

$\alpha\mu_1 + \beta\mu_2 \ll \mu_3$**:** Suppose $\mu_3(A) = 0$. Then $\mu_1(A) = \mu_2(A) = 0$, so $(\alpha\mu_1 + \beta\mu_2)(A) = 0$.

The derivative: For $A \in \mathcal{F}$, by Corollary 2.3.3,

$$(\alpha\mu_1 + \beta\mu_2)(A) = \alpha\int_A \frac{d\mu_1}{d\mu_3}\,d\mu_3 + \beta\int_A \frac{d\mu_2}{d\mu_3}\,d\mu_3 = \int_A \left(\alpha\frac{d\mu_1}{d\mu_3} + \beta\frac{d\mu_2}{d\mu_3}\right)d\mu_3.$$

Since $\alpha\frac{d\mu_1}{d\mu_3} + \beta\frac{d\mu_2}{d\mu_3}$ is nonnegative and measurable, it is a Radon-Nikodym derivative of $\alpha\mu_1 + \beta\mu_2$ w.r.t. μ_3, and by uniqueness the identity holds a.e. (μ_3). //

(iii)

If $\mu \ll \nu$ and $\frac{d\mu}{d\nu} > 0$ a.e. (ν), then $\nu \ll \mu$ and

$$\frac{d\nu}{d\mu} = \left(\frac{d\mu}{d\nu}\right)^{-1} \quad \text{a.e. } (\mu).$$

Answer

Proof. Write $f \equiv \frac{d\mu}{d\nu}$ and $N \equiv \{f = 0\}$, so that $\nu(N) = 0$. (Since μ is σ-finite, $f < \infty$ a.e. (ν), so we may take f finite-valued.) Define $g \equiv \frac{1}{f}I_{N^c}$, which is nonnegative and measurable (as in Problem 2.3), with $gf = I_{N^c}$.

Let $A \in \mathcal{F}$ be given. Since $\nu(N) = 0$, and by (1) with $h = I_A g$,

$$\nu(A) = \nu(A \cap N^c) = \int I_A\, gf\, d\nu = \int I_A\, g\, d\mu = \int_A g\, d\mu.$$

Thus $\nu \ll \mu$ (Example 4.1.3) and $\frac{d\nu}{d\mu} = g$ a.e. (μ) by uniqueness. Finally, $\mu(N) = \int_N f\, d\nu = 0$, since $f = 0$ on N. So $g = \frac{1}{f}$ a.e. (μ), i.e., $\frac{d\nu}{d\mu} = \left(\frac{d\mu}{d\nu}\right)^{-1}$ a.e. (μ). //

(iv)

Let $\{\mu_n\}_{n\geq 1}$ be a sequence of measures and $\{\alpha_n\}_{n\geq 1}$ be a sequence of positive real numbers, i.e., $\alpha_n > 0$ for all $n \geq 1$. Define $\mu = \sum_{n=1}^{\infty}\alpha_n\mu_n$.

(a) Then, $\mu \ll \nu$ iff $\mu_n \ll \nu$ for each $n \geq 1$ and in this case,

$$\frac{d\mu}{d\nu} = \sum_{n=1}^{\infty}\alpha_n\frac{d\mu_n}{d\nu} \quad \text{a.e. } (\nu).$$

(b) $\mu \perp \nu$ iff $\mu_n \perp \nu$ for all $n \geq 1$.

Answer

Proof. First, μ is a measure. Clearly $\mu(\emptyset) = 0$ and $\mu \geq 0$. For disjoint $A_1, A_2, \ldots \in \mathcal{F}$, by Problem 2.20,

$$\mu\Big(\bigcup_{k\geq 1} A_k\Big) = \sum_{n=1}^{\infty} \alpha_n \sum_{k=1}^{\infty} \mu_n(A_k) = \sum_{k=1}^{\infty}\sum_{n=1}^{\infty} \alpha_n \mu_n(A_k) = \sum_{k=1}^{\infty} \mu(A_k).$$

For any $A \in \mathcal{F}$, since every $\alpha_n > 0$ and every $\mu_n(A) \geq 0$,

$$\mu(A) = 0 \iff \mu_n(A) = 0 \text{ for all } n \geq 1. \tag{2}$$

(a):

($\Rightarrow$) Suppose $\mu \ll \nu$, and let $\nu(A) = 0$. Then $\mu(A) = 0$, so $\mu_n(A) = 0$ for all n by (2). Thus $\mu_n \ll \nu$ for each $n \geq 1$.

($\Leftarrow$) Suppose $\mu_n \ll \nu$ for each $n \geq 1$, and let $\nu(A) = 0$. Then $\mu_n(A) = 0$ for all n, so $\mu(A) = 0$ by (2). Thus $\mu \ll \nu$.

The derivative: In this case, for $A \in \mathcal{F}$, Corollary 2.3.5 gives

$$\mu(A) = \sum_{n=1}^{\infty} \alpha_n \int_A \frac{d\mu_n}{d\nu}\, d\nu = \int_A \sum_{n=1}^{\infty} \alpha_n \frac{d\mu_n}{d\nu}\, d\nu.$$

Since $\sum_{n=1}^{\infty} \alpha_n \frac{d\mu_n}{d\nu}$ is nonnegative and measurable, it is a Radon-Nikodym derivative of μ w.r.t. ν, and by uniqueness the identity holds a.e. (ν).

(b):

($\Rightarrow$) Suppose $\mu \perp \nu$, with $B \in \mathcal{F}$ such that $\mu(B) = 0$ and $\nu(B^c) = 0$. By (2), $\mu_n(B) = 0$ for all n, so the same B gives $\mu_n \perp \nu$ for all $n \geq 1$.

($\Leftarrow$) Suppose $\mu_n \perp \nu$ for all $n \geq 1$, with $B_n \in \mathcal{F}$ such that $\mu_n(B_n) = 0$ and $\nu(B_n^c) = 0$. Define $D \equiv \bigcap_{n\geq 1} B_n \in \mathcal{F}$. Since $D \subset B_n$, $\mu_n(D) = 0$ for all n, so $\mu(D) = 0$ by (2). Since $D^c = \bigcup_{n\geq 1} B_n^c$, countable subadditivity gives $\nu(D^c) \leq \sum_{n=1}^{\infty} \nu(B_n^c) = 0$. Thus $\mu \perp \nu$. //

Q3: 4.5

Find the Lebesgue decomposition of μ w.r.t. ν and the Radon-Nikodym derivative $\frac{d\mu_a}{d\nu}$ in the following cases where μ_a is the absolutely continuous component of μ w.r.t. ν.

The measures referred to in (a)–(f) are defined in Tables 4.6.1 and 4.6.2, given at the end of this section.

Answer

Note: Table 4.6.1 prints Geometric(p) as $\sum_{i=0}^{\infty} p(1-p)^{i-1} I_A(i)$, which does not sum to 1. We take Geometric(p) on $\mathbb{N} = \{1, 2, \ldots\}$ with mass $p(1-p)^{i-1}$ at i. In particular, Geometric(p) puts no mass at 0.

We utilize the following results:

- Definitions 4.1.1–4.1.3, Example 4.1.3, and Theorem 4.1.1 (see Q2).
- Proposition 4.1.2 (see Q2), and the Lemma (1).
- Corollary 2.3.3 (see Q1).

Write m for Lebesgue measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ and κ for the counting measure on $\mathbb{Z}_+ = \{0, 1, 2, \ldots\}$, i.e., $\kappa(A) \equiv \#(A \cap \mathbb{Z}_+)$, so that $\int_A h\, d\kappa = \sum_{k\in A\cap\mathbb{Z}_+} h(k)$ for nonnegative h (as in Remark 2.3.2). From Tables 4.6.1 and 4.6.2, the measures here have the following densities.

Measure	Density	w.r.t.
$N(0,1)$	$\phi(x) \equiv \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$	m
Exponential(1)	$e^{-x} I_{(0,\infty)}(x)$	m
Cauchy$(0,1)$	$\psi(x) \equiv \frac{1}{\pi(1+x^2)}$	m
Poisson(λ)	$e^{-\lambda} \frac{\lambda^x}{x!} I_{\mathbb{Z}_+}(x)$	κ
Geometric(p)	$p(1-p)^{x-1} I_{\mathbb{N}}(x)$	κ
Binomial$(10, 1/2)$	$\binom{10}{x} 2^{-10} I_{\{0,1,\ldots,10\}}(x)$	κ

Any measure with a density w.r.t. m vanishes on countable sets, since it is $\ll m$ (Example 4.1.3) and m of a countable set is 0.

Claim: Let η be a measure, and let μ and ν be finite measures with $\mu(A) = \int_A f \, d\eta$ and $\nu(A) = \int_A g \, d\eta$ for all A, for nonnegative measurable f, g. Let $E \equiv \{g > 0\}$. Then the Lebesgue decomposition of μ w.r.t. ν is

$$\mu_a(A) = \mu(A \cap E), \quad \mu_s(A) = \mu(A \cap E^c), \quad \text{with} \quad \frac{d\mu_a}{d\nu} = \frac{f}{g} I_E \ \text{ a.e. } (\nu). \tag{3}$$

Proof. μ_a and μ_s are finite measures (restrictions of μ) with $\mu = \mu_a + \mu_s$. $\mu_s \perp \nu$, since $\mu_s(E) = 0$ and $\nu(E^c) = \int_{E^c} g \, d\eta = 0$, as $g = 0$ on E^c. Next, $\frac{f}{g} I_E$ is nonnegative and measurable (as in Problem 2.3), and $\frac{f}{g} I_E \cdot g = f I_E$. So by (1), applied to ν and η with $h = I_A \frac{f}{g} I_E$,

$$\int_A \frac{f}{g} I_E \, d\nu = \int I_A \frac{f}{g} I_E \, g \, d\eta = \int_{A \cap E} f \, d\eta = \mu_a(A).$$

Thus $\mu_a \ll \nu$ (Example 4.1.3) with $\frac{d\mu_a}{d\nu} = \frac{f}{g} I_E$. By the uniqueness in Theorem 4.1.1(i), (μ_a, μ_s) is the Lebesgue decomposition. //

(a)

$\mu = N(0,1)$, $\nu =$ Exponential(1)

Answer

$\mu_a(A) = \mu(A \cap (0,\infty))$, $\mu_s(A) = \mu(A \cap (-\infty, 0])$, and

$$\frac{d\mu_a}{d\nu}(x) = \frac{1}{\sqrt{2\pi}} e^{x - x^2/2} I_{(0,\infty)}(x) \quad \text{a.e. } (\nu).$$

Proof. Apply (3) with $\eta = m$, $f = \phi$, and $g(x) = e^{-x} I_{(0,\infty)}(x)$, so that $E = (0,\infty)$ and $\frac{f}{g} I_E = \frac{1}{\sqrt{2\pi}} e^{-x^2/2} e^x I_{(0,\infty)}$. //

(b)

$\mu =$ Exponential(1), $\nu = N(0,1)$

Answer

$\mu_a = \mu$ and $\mu_s = 0$, i.e., $\mu \ll \nu$, and

$$\frac{d\mu}{d\nu}(x) = \sqrt{2\pi}\, e^{x^2/2 - x} I_{(0,\infty)}(x) \quad \text{a.e. } (\nu).$$

Proof. Apply (3) with $\eta = m$, $f(x) = e^{-x} I_{(0,\infty)}(x)$, and $g = \phi > 0$, so that $E = \mathbb{R}$, $\mu_s(A) = \mu(A \cap \emptyset) = 0$, and $\frac{f}{g} = \sqrt{2\pi}\, e^{x^2/2} e^{-x} I_{(0,\infty)}$. //

(c)

$\mu = \mu_1 + \mu_2$, where $\mu_1 = N(0,1)$, $\mu_2 = \text{Poisson}(1)$ and $\nu = \text{Cauchy}(0,1)$.

Answer

$\mu_a = \mu_1$ and $\mu_s = \mu_2$, and

$$\frac{d\mu_a}{d\nu}(x) = \sqrt{\frac{\pi}{2}}\,(1+x^2)\,e^{-x^2/2} \quad \text{a.e. } (\nu).$$

Proof. $\mu_1 \ll \nu$**:** Apply (3) to μ_1 and ν with $\eta = m$, $f = \phi$, and $g = \psi > 0$. Then $E = \mathbb{R}$, so $\mu_1 \ll \nu$ with $\frac{d\mu_1}{d\nu} = \frac{\phi}{\psi} = \sqrt{\frac{\pi}{2}}(1+x^2)e^{-x^2/2}$.

$\mu_2 \perp \nu$**:** Let $D \equiv \mathbb{R} \setminus \mathbb{Z}_+$. Then $\mu_2(D) = 0$, and $\nu(D^c) = \nu(\mathbb{Z}_+) = 0$, since $\mathbb{Z}_+$ is countable and ν has a density w.r.t. m.

Putting this all together, $\mu = \mu_1 + \mu_2$ with $\mu_1 \ll \nu$ and $\mu_2 \perp \nu$, so by the uniqueness in Theorem 4.1.1(i), $\mu_a = \mu_1$, $\mu_s = \mu_2$, and $\frac{d\mu_a}{d\nu} = \frac{d\mu_1}{d\nu}$. //

(d)

$\mu = \mu_1 + \mu_2$, $\nu = \text{Geometric}(p)$, $0 < p < 1$, where $\mu_1 = N(0,1)$ and $\mu_2 = \text{Poisson}(1)$.

Answer

$\mu_a(A) = \mu_2(A \cap \mathbb{N})$, $\mu_s(A) = \mu_1(A) + \mu_2(A \cap \{0\})$, and

$$\frac{d\mu_a}{d\nu}(x) = \frac{(1-p)^{1-x}}{e\,p\,x!} I_{\mathbb{N}}(x) \quad \text{a.e. } (\nu).$$

Proof. μ_2 **against** ν**:** Apply (3) to μ_2 and ν with $\eta = \kappa$, $f(x) = \frac{e^{-1}}{x!} I_{\mathbb{Z}_+}(x)$, and $g(x) = p(1-p)^{x-1} I_{\mathbb{N}}(x)$, so that $E = \mathbb{N}$. Then $\mu_2 = \mu_{2a} + \mu_{2s}$, where $\mu_{2a}(A) \equiv \mu_2(A \cap \mathbb{N}) \ll \nu$ and $\mu_{2s}(A) \equiv \mu_2(A \cap \mathbb{N}^c) = \mu_2(A \cap \{0\})$, as μ_2 lives on $\mathbb{Z}_+$. Also

$$\frac{d\mu_{2a}}{d\nu}(x) = \frac{e^{-1}/x!}{p(1-p)^{x-1}} I_{\mathbb{N}}(x) = \frac{(1-p)^{1-x}}{e\,p\,x!} I_{\mathbb{N}}(x).$$

$\mu_1 + \mu_{2s} \perp \nu$**:** Take $D \equiv \mathbb{N}$. Then $\mu_1(\mathbb{N}) = 0$, since $\mathbb{N}$ is countable and μ_1 has a density w.r.t. m, and $\mu_{2s}(\mathbb{N}) = \mu_2(\mathbb{N} \cap \{0\}) = 0$. Also $\nu(\mathbb{N}^c) = 0$.

Putting this all together, $\mu = \mu_{2a} + (\mu_1 + \mu_{2s})$ with $\mu_{2a} \ll \nu$ and $\mu_1 + \mu_{2s} \perp \nu$, so by uniqueness, $\mu_a = \mu_{2a}$ and $\mu_s = \mu_1 + \mu_{2s}$. //

(e)

$\mu = \mu_1 + \mu_2$, $\nu = \nu_1 + \nu_2$ where $\mu_1 = N(0,1)$, $\mu_2 = \text{Poisson}(1)$, $\nu_1 = \text{Cauchy}(0,1)$ and $\nu_2 = \text{Geometric}(p)$, $0 < p < 1$.

Answer

$\mu_a(A) = \mu_1(A) + \mu_2(A \cap \mathbb{N})$, $\mu_s(A) = \mu_2(A \cap \{0\})$, and

$$\frac{d\mu_a}{d\nu}(x) = \sqrt{\frac{\pi}{2}}\,(1+x^2)\,e^{-x^2/2} I_{\mathbb{N}^c}(x) + \frac{(1-p)^{1-x}}{e\,p\,x!} I_{\mathbb{N}}(x) \quad \text{a.e. } (\nu).$$

Proof. Let μ_{2a} and μ_{2s} be as in (d). The tricky part is that ν adds mass on $\mathbb{N}$, so $\frac{d\mu_1}{d\nu}$ is not $\frac{d\mu_1}{d\nu_1}$ everywhere. We pass through ν_1 and ν_2.

$\nu_1, \nu_2 \ll \nu$**:** Note that $\nu_1(\mathbb{N}) = 0$, since ν_1 has a density w.r.t. m, and $\nu_2(\mathbb{N}^c) = 0$. So for $A \in \mathcal{B}(\mathbb{R})$,

$$\nu_1(A) = \nu_1(A \cap \mathbb{N}^c) = \nu(A \cap \mathbb{N}^c) = \int_A I_{\mathbb{N}^c}\, d\nu \quad \text{and} \quad \nu_2(A) = \nu_2(A \cap \mathbb{N}) = \nu(A \cap \mathbb{N}) = \int_A I_{\mathbb{N}}\, d\nu.$$

Thus $\nu_1, \nu_2 \ll \nu$, with $\frac{d\nu_1}{d\nu} = I_{\mathbb{N}^c}$ and $\frac{d\nu_2}{d\nu} = I_{\mathbb{N}}$.

$\mu_1 \ll \nu$**:** By (c), $\mu_1 \ll \nu_1$ with $\frac{d\mu_1}{d\nu_1} = \frac{\phi}{\psi}$. By Proposition 4.1.2(i),

$$\frac{d\mu_1}{d\nu} = \frac{d\mu_1}{d\nu_1}\frac{d\nu_1}{d\nu} = \sqrt{\frac{\pi}{2}}(1 + x^2)e^{-x^2/2} I_{\mathbb{N}^c}(x) \quad \text{a.e. } (\nu).$$

$\mu_{2a} \ll \nu$**:** By (d), with ν_2 in the role of ν, $\mu_{2a} \ll \nu_2$ with $\frac{d\mu_{2a}}{d\nu_2} = \frac{(1-p)^{1-x}}{e\,p\,x!} I_{\mathbb{N}}$. By Proposition 4.1.2(i),

$$\frac{d\mu_{2a}}{d\nu} = \frac{d\mu_{2a}}{d\nu_2}\frac{d\nu_2}{d\nu} = \frac{(1-p)^{1-x}}{e\,p\,x!} I_{\mathbb{N}}(x) \quad \text{a.e. } (\nu).$$

$\mu_{2s} \perp \nu$**:** Let $D \equiv \mathbb{R} \setminus \{0\}$. Then $\mu_{2s}(D) = 0$, and $\nu(\{0\}) = \nu_1(\{0\}) + \nu_2(\{0\}) = 0 + 0 = 0$.

Putting this all together, $\mu = (\mu_1 + \mu_{2a}) + \mu_{2s}$, where $\mu_1 + \mu_{2a} \ll \nu$ by Proposition 4.1.2(ii) and $\mu_{2s} \perp \nu$. By uniqueness, $\mu_a = \mu_1 + \mu_{2a}$ and $\mu_s = \mu_{2s}$, and Proposition 4.1.2(ii) gives $\frac{d\mu_a}{d\nu} = \frac{d\mu_1}{d\nu} + \frac{d\mu_{2a}}{d\nu}$ a.e. (ν). //

(f)

$\mu = \text{Binomial}(10, 1/2)$, $\nu = \text{Poisson}(1)$.

Answer

$\mu_a = \mu$ and $\mu_s = 0$, i.e., $\mu \ll \nu$, and

$$\frac{d\mu}{d\nu}(x) = \binom{10}{x}\frac{e\,x!}{2^{10}} I_{\{0,\ldots,10\}}(x) = \frac{e \cdot 10!}{2^{10}\,(10-x)!} I_{\{0,\ldots,10\}}(x) \quad \text{a.e. } (\nu).$$

Proof. Apply (3) with $\eta = \kappa$, $f(x) = \binom{10}{x}2^{-10} I_{\{0,\ldots,10\}}(x)$, and $g(x) = \frac{e^{-1}}{x!} I_{\mathbb{Z}_+}(x)$. Then $E = \mathbb{Z}_+ \supset \{0, \ldots, 10\}$, so $\mu_s(A) = \mu(A \cap \mathbb{Z}_+^c) = 0$, and $\frac{f}{g} I_E$ is as displayed, using $\binom{10}{x} x! = \frac{10!}{(10-x)!}$. //

Q4: Extra A

Let $(\Omega, \mathcal{F})$ be a measurable space, and $\mu, \mu_1, \mu_2, \nu, \rho$ are σ-finite measures on it. Fix a constant $c > 0$ and a set $B \in \mathcal{F}$, and define the set function $\lambda(A) \equiv c\,\mu_1(A \cap B)$ for all $A \in \mathcal{F}$. Then, prove the following:

Answer

We utilize the following results:

- Definitions 4.1.1–4.1.3, Example 4.1.3, and Theorem 4.1.1 (see Q2).
- Corollary 2.3.3 (see Q1).
- The Claim (3), with m and κ as in Q3.

(a)

λ is a σ-finite measure.

Answer

Proof. λ **is a measure:** Clearly $\lambda(\emptyset) = c\,\mu_1(\emptyset) = 0$ and $\lambda \geq 0$, as $c > 0$. For disjoint $A_1, A_2, \ldots \in \mathcal{F}$, the sets $A_k \cap B$ are disjoint with union $\big(\bigcup_{k\geq 1} A_k\big) \cap B$, so

$$\lambda\Big(\bigcup_{k\geq 1} A_k\Big) = c\sum_{k=1}^{\infty} \mu_1(A_k \cap B) = \sum_{k=1}^{\infty} \lambda(A_k).$$

λ **is σ-finite:** Take a partition $\{D_n\}$ of Ω with $\mu_1(D_n) < \infty$. Then $\lambda(D_n) = c\,\mu_1(D_n \cap B) \leq c\,\mu_1(D_n) < \infty$ for all $n \geq 1$. //

(b)

If $\mu_1 \ll \nu$, then $\lambda \ll \nu$.

Answer

Proof. Suppose $\nu(A) = 0$ for some $A \in \mathcal{F}$. Then $\nu(A \cap B) = 0$, since $A \cap B \subset A$, and so $\mu_1(A \cap B) = 0$, since $\mu_1 \ll \nu$. Thus $\lambda(A) = c \cdot 0 = 0$, and $\lambda \ll \nu$. //

(c)

Express the Radon-Nikodym derivative $\frac{d\lambda}{d\nu}$ in terms of $\frac{d\mu_1}{d\nu}$ (proof required).

Answer

$\frac{d\lambda}{d\nu} = c\,I_B \frac{d\mu_1}{d\nu}$ a.e. (ν).

Proof. Here $\mu_1 \ll \nu$, as in (b), so $\frac{d\mu_1}{d\nu}$ exists by Theorem 4.1.1. For $A \in \mathcal{F}$, since $I_{A\cap B} = I_A I_B$, Corollary 2.3.3 gives

$$\lambda(A) = c\int_{A\cap B} \frac{d\mu_1}{d\nu}\,d\nu = \int_A c\,I_B \frac{d\mu_1}{d\nu}\,d\nu.$$

Since $c\,I_B \frac{d\mu_1}{d\nu}$ is nonnegative and measurable, it is a Radon-Nikodym derivative of λ w.r.t. ν, and by uniqueness the identity holds a.e. (ν), as λ is σ-finite by (a). //

(d)

If $\mu \perp \nu$ and $\rho \ll \nu$, show that $\mu \perp \rho$.

Answer

Proof. Suppose $\mu \perp \nu$, with $D \in \mathcal{F}$ such that $\mu(D) = 0$ and $\nu(D^c) = 0$ (we write D, as B is fixed above). Since $\rho \ll \nu$, $\rho(D^c) = 0$, so the same D gives $\mu \perp \rho$. //

(e)

Let $(\Omega, \mathcal{F}) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$, $\mu_1 =$ Exponential(2) (rate parameterization), $c = 3$, $B = [1, 8]$, $\mu_2 =$ Binomial$(10, 0.4)$ and $\nu =$ Uniform$(0, 6)$. Define $\mu_3 = \lambda + \mu_2$. Find the Lebesgue decomposition of μ_3 w.r.t. ν, and find the Radon-Nikodym derivative $\frac{d\mu_{3a}}{d\nu}$, where μ_{3a} is the absolutely continuous component of μ_3 in the decomposition.

Answer

$\mu_{3a}(A) = 3\mu_1(A \cap [1,6])$, $\mu_{3s}(A) = 3\mu_1(A \cap (6,8]) + \mu_2(A)$, and

$$\frac{d\mu_{3a}}{d\nu}(x) = 36\, e^{-2x} I_{[1,6]}(x) \quad \text{a.e. } (\nu).$$

Note: Table 4.6.2 parameterizes Exponential(β) by scale, with density $\frac{1}{\beta} e^{-x/\beta} I_{(0,\infty)}(x)$. So Exponential(2) in the rate parameterization is the table's Exponential(1/2). From Tables 4.6.1 and 4.6.2, the measures here have the following densities.

Measure	Density	w.r.t.
Exponential(2) (rate)	$2e^{-2x} I_{(0,\infty)}(x)$	m
Uniform$(0,6)$	$\frac{1}{6} I_{[0,6]}(x)$	m
Binomial$(10, 0.4)$	$\binom{10}{x}(0.4)^x (0.6)^{10-x} I_{\{0,1,\ldots,10\}}(x)$	κ

Proof. The tricky part is that $B = [1,8]$ overhangs $[0,6]$, where ν lives, so λ itself splits. We decompose λ first.

Density of λ: By (c), with m in the role of ν (as $\mu_1 \ll m$),

$$\frac{d\lambda}{dm}(x) = 3I_{[1,8]}(x) \cdot 2e^{-2x} I_{(0,\infty)}(x) = 6e^{-2x} I_{[1,8]}(x).$$

Also, λ is finite, as $\lambda(\mathbb{R}) \le 3\mu_1(\mathbb{R}) = 3$.

λ against ν: Apply (3) to λ and ν with $\eta = m$, $f(x) = 6e^{-2x} I_{[1,8]}(x)$, and $g(x) = \frac{1}{6} I_{[0,6]}(x)$, so that $E = [0,6]$. Then $\lambda = \lambda_a + \lambda_s$, where $\lambda_a(A) \equiv \lambda(A \cap [0,6]) = 3\mu_1(A \cap [1,6]) \ll \nu$ and $\lambda_s(A) \equiv \lambda(A \cap [0,6]^c) = 3\mu_1(A \cap (6,8])$. Also

$$\frac{d\lambda_a}{d\nu}(x) = \frac{6e^{-2x} I_{[1,8]}(x)}{1/6} I_{[0,6]}(x) = 36\, e^{-2x} I_{[1,6]}(x).$$

$\lambda_s + \mu_2 \perp \nu$: Take $D \equiv [0,6] \setminus \{0,1,\ldots,10\}$. Then $\lambda_s(D) = 0$, since $D \subset [0,6]$, and $\mu_2(D) = 0$, since $D \cap \mathbb{Z}_+ = \emptyset$. Also $\nu(D^c) \le \nu([0,6]^c) + \nu(\{0,1,\ldots,10\}) = 0$, since $g = 0$ off $[0,6]$, and $\{0,1,\ldots,10\}$ is countable and ν has a density w.r.t. m.

Putting this all together, $\mu_3 = \lambda_a + (\lambda_s + \mu_2)$ with $\lambda_a \ll \nu$ and $\lambda_s + \mu_2 \perp \nu$, so by the uniqueness in Theorem 4.1.1(i), $\mu_{3a} = \lambda_a$, $\mu_{3s} = \lambda_s + \mu_2$, and $\frac{d\mu_{3a}}{d\nu} = \frac{d\lambda_a}{d\nu}$. //